

NATURAL LANGUAGE PROCESSING

PRACTICAL EXERCISE – 1

Perform word level analysis on a user input text using NLP

SOURCE CODE :-

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer

from collections import Counter


# Download required resources (run once)

nltk.download('punkt')

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('punkt_tab') # Added punkt_tab download


# Get input from the user

text = input("Enter a sentence: ")


# Step 1: Tokenization

tokens = word_tokenize(text)

print("\nTokens:", tokens)


# Step 2: Total number of words

print("Total Words:", len(tokens))


# Step 3: Stop-word removal
```

```
stop_words = set(stopwords.words('english'))  
filtered_words = [word for word in tokens if word.lower() not in stop_words]
```

```
print("Words after removing stop words:", filtered_words)
```

```
# Step 4: Word frequency
```

```
frequency = Counter(filtered_words)
```

```
print("Word Frequency:", frequency)
```

```
# Step 5: Stemming
```

```
stemmer = PorterStemmer()
```

```
stems = [stemmer.stem(word) for word in filtered_words]
```

```
print("Stemmed Words:", stems)
```

```
# Step 6: Lemmatization
```

```
lemmatizer = WordNetLemmatizer()
```

```
lemmas = [lemmatizer.lemmatize(word) for word in filtered_words]
```

```
print("Lemmatized Words:", lemmas)
```